6 Months Training TR-104 Full Stack Development

January, 2026

Day 12 – Advanced Validation and Authentication Service

1. Objective of the Day

The objective was to enhance the authentication system by implementing advanced validation and designing a centralized authentication service using dependency injection.

2. Tasks Performed

- Enhanced validation rules for username, email, and password fields
- Implemented minimum length and format-based validation
- Created an authentication service to manage login state
- Stored user details such as username, email, and role
- Implemented mock logic for assigning user roles
- Used dependency injection to integrate the service into components
- Exposed authentication data using getter methods
- Studied TypeScript strict mode behavior in templates
- Resolved issues related to form control access using bracket notation

3. Tools / Technologies Used

- Angular Framework
- TypeScript
- Reactive Forms
- Angular Services

4. Key Learnings

- Validation improves data accuracy and user experience
- Services act as a central source of application state
- Dependency Injection enables efficient sharing of services
- Getter methods provide controlled access to data
- TypeScript strict mode enforces safer coding practices

5. Challenges Faced

- Debugging TypeScript strict mode errors in templates
- Implementing validation logic correctly

6. Solutions

- Used bracket notation to access form controls safely
- Practiced validation rules with multiple test cases

7. Conclusion

- Successfully enhanced form validation and authentication logic
- Established a centralized authentication service